

Information Field Theory

And Evolutionary Biology

IFT-Evolution v1.0

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The application of Information Field Theory (IFT) to evolutionary biology is presented. Evolution by natural selection is reinterpreted as an informational optimization process: the fitness of an organism is the KL divergence between its phenotypic distribution ρ and the adaptive optimum ρ^* . Speciation occurs as an informational phase transition when the entropy of the adaptive landscape changes discontinuously. The evolution of functional quantum coherence (IFT-Bio document) is explained as the result of natural selection on the fractal dimension D_f : organisms with higher D_f (super-Ohmic baths) better preserve coherence and gain an adaptive advantage. An informational replicator-mutator equation is derived that generalizes standard evolutionary dynamics. Testable predictions in experimental evolution with bacteria and viruses are discussed.

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Chapter 1

Introduction: Evolution as Informational Optimization

1.1 Motivation

The synthetic theory of evolution (neo-Darwinism) explains evolutionary change through random genetic variation and natural selection. However, it lacks a formal framework for quantifying the “information” that natural selection accumulates. IFT fills this gap: evolution is a process of optimization of the informational field ρ under environmental constraints.

1.2 Central Thesis

Theorem 1.2.1 (IFT-Evolution Thesis). *Natural selection is an informational optimization algorithm that maximizes fitness understood as the reduction of the KL divergence between the population phenotypic distribution ρ and the optimal distribution ρ^* imposed by the environment. Speciation is an informational phase transition, and biological quantum coherence evolved by natural selection on the fractal dimension D_f .*

Chapter 2

Fitness as Informational Divergence

2.1 Definition of Informational Fitness

Definition 2.1.1 (Fitness as KL Divergence). *For a population with phenotypic distribution $\rho(x)$ (where x is a vector of traits) and an environmental optimum $\rho^*(x)$:*

$$F[\rho] = -D_{\text{KL}}(\rho||\rho^*) = - \int \rho(x) \ln \frac{\rho(x)}{\rho^*(x)} dx. \quad (2.1)$$

The population maximizes $F[\rho]$ (minimizes the divergence) over the course of evolution.

2.2 Properties

1. $F[\rho] \leq 0$ (the KL divergence is non-negative).
2. $F[\rho] = 0$ only if $\rho = \rho^*$ (perfect adaptation).
3. The rate of change of fitness is:

$$\frac{dF}{dt} = \int \partial_t \rho(x) \left(\ln \frac{\rho^*(x)}{\rho(x)} - 1 \right) dx. \quad (2.2)$$

2.3 Informational Replicator-Mutator Equation

$$\boxed{\partial_t \rho(x) = \rho(x) (f(x) - \bar{f}) + \mu \nabla^2 \rho(x)}, \quad (2.3)$$

where $f(x)$ is the individual fitness of phenotype x , $\bar{f} = \int f(x) \rho(x) dx$ is the mean fitness, and μ is the mutation rate (diffusion in phenotypic space). This equation generalizes the standard replicator dynamics and the Price equation.

Chapter 3

Speciation as an Informational Phase Transition

3.1 Adaptive Landscape as Free Energy

The adaptive landscape of a population is defined as:

$$\mathcal{F}[\rho] = -F[\rho] - T_{\text{evol}}S[\rho], \quad (3.1)$$

where T_{evol} is an “evolutionary temperature” measuring the intensity of genetic drift, and $S[\rho] = -\int \rho \ln \rho dx$ is the phenotypic entropy.

3.2 Phase Transition

Speciation occurs when $\mathcal{F}[\rho]$ develops multiple local minima separated by free-energy barriers. The population that previously occupied a single adaptive peak splits into two subpopulations occupying distinct peaks.

Theorem 3.2.1 (Speciation Condition). *Speciation occurs when the Hessian matrix of $\mathcal{F}$ at equilibrium develops a negative eigenvalue:*

$$\det(\nabla^2 \mathcal{F}[\rho_{\text{eq}}]) < 0. \quad (3.2)$$

This corresponds to a bifurcation in the replicator-mutator dynamics.

Chapter 4

Evolution of Functional Quantum Coherence

4.1 Quantum Coherence as an Adaptive Trait

The IFT-Bio document demonstrated that functional quantum coherence depends on the fractal dimension D_f (super-Ohmic baths protect coherence). IFT-Evolution explains how D_f evolves by natural selection:

1. Organisms with higher D_f (more complex and fractal molecular structure) have super-Ohmic baths ($s = 1 + D_f \geq 3$).
2. These baths suppress decoherence, enabling functional quantum processes (more efficient photosynthesis, faster enzymatic catalysis).
3. The adaptive advantage translates into greater differential fitness.
4. Over generations, D_f increases until reaching an optimal value limited by biochemical constraints.

4.2 Evolution Equation for D_f

$$\frac{dD_f}{dt} = \sigma_D^2 \frac{\partial \bar{f}}{\partial D_f}, \quad (4.1)$$

where σ_D^2 is the genetic variance of D_f in the population and $\partial \bar{f} / \partial D_f$ is the selection gradient.

4.3 Testable Prediction

In controlled evolution experiments with photosynthetic bacteria, D_f (measurable via fractal analysis of protein structure) is predicted to increase over generations if greater photosynthetic efficiency is selected for.

Chapter 5

Entropy and Evolutionary Complexity

5.1 Growth of Complexity

IFT predicts that phenotypic entropy $S[\rho]$ tends to increase over the course of evolution (generalized H-theorem), but adaptation ($F[\rho] \rightarrow 0$) can reduce entropy locally. The balance between the two determines the evolutionary trajectory.

Theorem 5.1.1 (Evolutionary Second Law).

$$\frac{d}{dt} (S[\rho] - \beta F[\rho]) \geq 0, \quad (5.1)$$

where β is a parameter weighting the relative importance of adaptation versus diversification.

5.2 Interpretation

Evolution tends to increase the total “informational complexity”, defined as $S[\rho] - \beta F[\rho]$. This unifies the observation that life tends to diversify (increase in entropy) and to adapt (reduction in KL divergence).

Chapter 6

Comparison with Existing Evolutionary Theories

Concept	Neo-Darwinism	IFT-Evolution
Fitness	Reproductive (number of offspring)	Informational (KL divergence)
Natural selection	Filter of genetic variants	Optimization of $\mathcal{F}[\rho]$
Speciation	Reproductive isolation	Informational phase transition
Adaptation	Increase in reproductive fitness	$\rho \rightarrow \rho^*$ (reduction of KL)
Genetic drift	Neutral stochastic process	Thermal fluctuations in $\mathcal{F}$
Quantum coherence	Not explained	Selection on D_f

Chapter 7

Testable Predictions

1. **Evolution of D_f :** In long-term evolution experiments (Lenski, *E. coli*), measure D_f of proteins involved in energy metabolism. Prediction: D_f will increase in lineages with higher fitness.
2. **Experimental speciation:** In yeast or bacterial populations subjected to heterogeneous environments, predict the moment of speciation when $\det(\nabla^2 \mathcal{F}) < 0$.
3. **Replicator-mutator equation:** Fit experimental evolution data to the IFT equation and compare with standard models (prediction: better fit with the informational diffusion term).

Chapter 8

Master Formula of IFT-Evolution

$$F[\rho] = -D_{\text{KL}}(\rho||\rho^*) \quad (\text{Informational fitness}), \quad (8.1)$$

$$\partial_t \rho = \rho(f - \bar{f}) + \mu \nabla^2 \rho \quad (\text{Replicator-mutator}), \quad (8.2)$$

$$\mathcal{F}[\rho] = -F[\rho] - T_{\text{evol}} S[\rho] \quad (\text{Evolutionary free energy}), \quad (8.3)$$

$$\text{Speciation: } \det(\nabla^2 \mathcal{F}) < 0 \quad (\text{Phase transition}), \quad (8.4)$$

$$\frac{dD_f}{dt} = \sigma_D^2 \frac{\partial \bar{f}}{\partial D_f} \quad (\text{Evolution of coherence}). \quad (8.5)$$

Chapter 9

Final Declaration

IFT-Evolution unifies evolutionary biology with the informational framework. Natural selection is optimization of the KL divergence, speciation is a phase transition, and quantum coherence evolved by selection on D_f . The framework provides quantitative predictions that are testable in experimental evolution.

$$\boxed{\text{Evolution} = \text{Informational optimization of } \rho \rightarrow \rho^*}.$$

$$\boxed{\text{Fitness} = -\text{KL divergence}(\rho||\rho^*)}.$$

$$\boxed{\text{Speciation} = \text{Phase transition in } \mathcal{F}[\rho]}.$$

9.1 Affirmations

1. Fitness is an informational measure: KL divergence between phenotype and optimum.
2. Natural selection optimizes the evolutionary free energy $\mathcal{F}[\rho]$.
3. Speciation is an informational phase transition.
4. Functional quantum coherence evolved by natural selection on D_f .
5. Evolutionary complexity increases according to the generalized second law.

9.2 Open Problems

1. Measure $\rho(x)$ in real populations from phenotypic and genomic data.
2. Estimate T_{evol} (evolutionary temperature) in different lineages.
3. Experimentally verify the informational replicator-mutator equation.
4. Test the speciation-as-bifurcation prediction against field data.
5. Measure D_f in proteins along experimental evolution lines.